

Please check the examination details below before entering your candidate information

Candidate surname	Other names
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Pearson
Edexcel Award

Centre Number

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 Candidate Number

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Tuesday 20 October 2020

Afternoon (Time: 3 hours)	Paper Reference 9811/01
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Advanced Extension Award
Mathematics

You must have: Mathematical Formulae and Statistical Tables An insert for questions 3, 4, 5 and 6	Total Marks
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Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible, e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1)** or **(+S2)**.
- There are 6 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

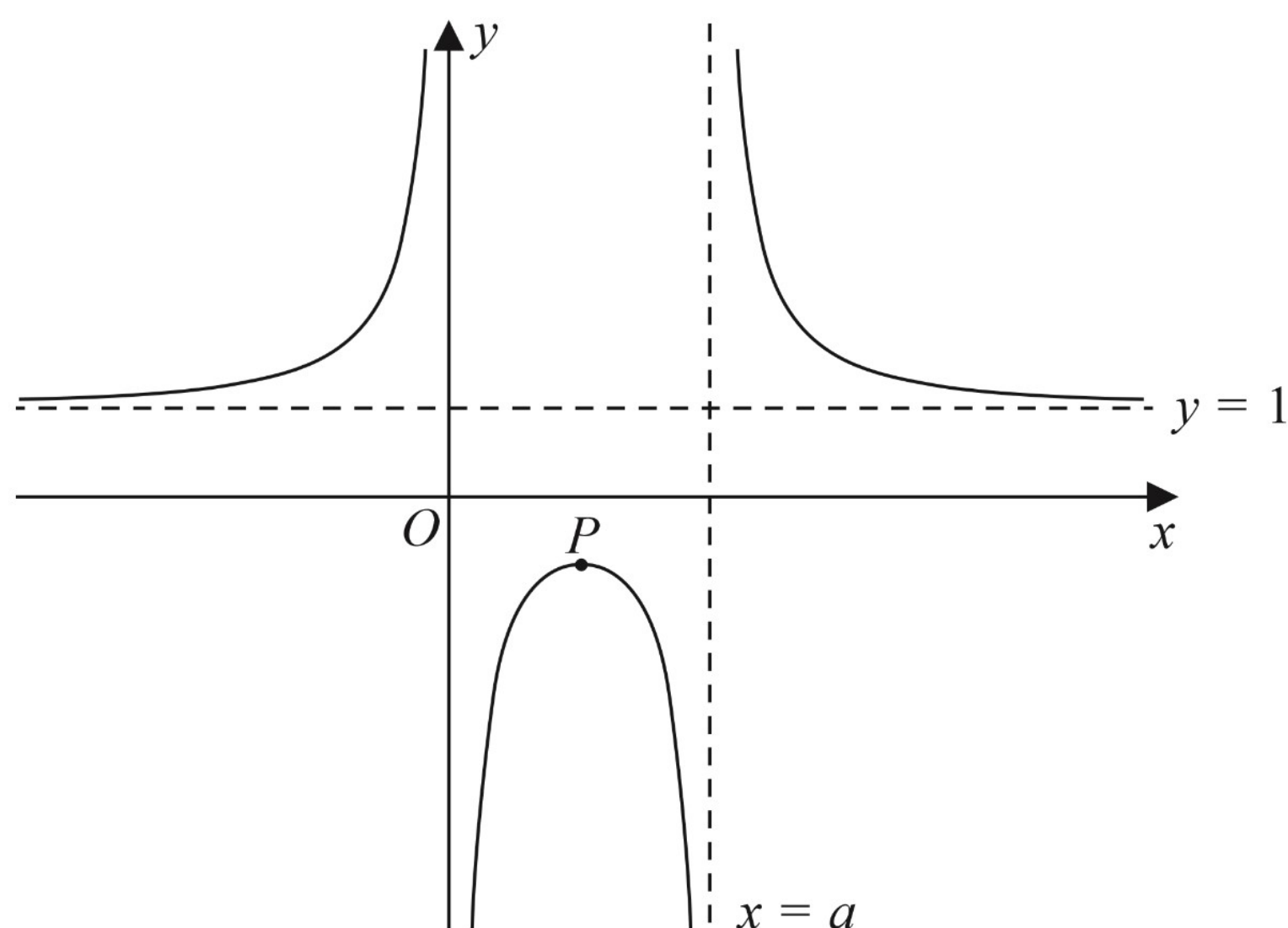


Figure 1

Figure 1 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = 1 + \frac{4}{x(x-3)}$$

The curve has a turning point at the point P , and the lines with equations $y = 1$, $x = 0$ and $x = a$ are asymptotes to the curve.

(a) Write down the value of a . (1)

(b) Find the coordinates of P , justifying your answer. (4)

(c) Sketch the curve with equation $y = \left| f\left(x + \frac{3}{2}\right) \right| - 1$

On your sketch, you should show the coordinates of any points of intersection with the coordinate axes, the coordinates of any turning points and the equations of any asymptotes.

(7)

(a) $x = 3$ IS EQUATION OF ASYMPTOTE

$$\therefore a = 3$$

(b) $f(x) = 1 + \frac{4}{x^2 - 3x}$

$$f'(x) = \frac{(x^2 - 3x)(0) - (4)(2x - 3)}{(x^2 - 3x)^2} = \frac{-4(2x - 3)}{(x^2 - 3x)^2}$$



Question 1 continued

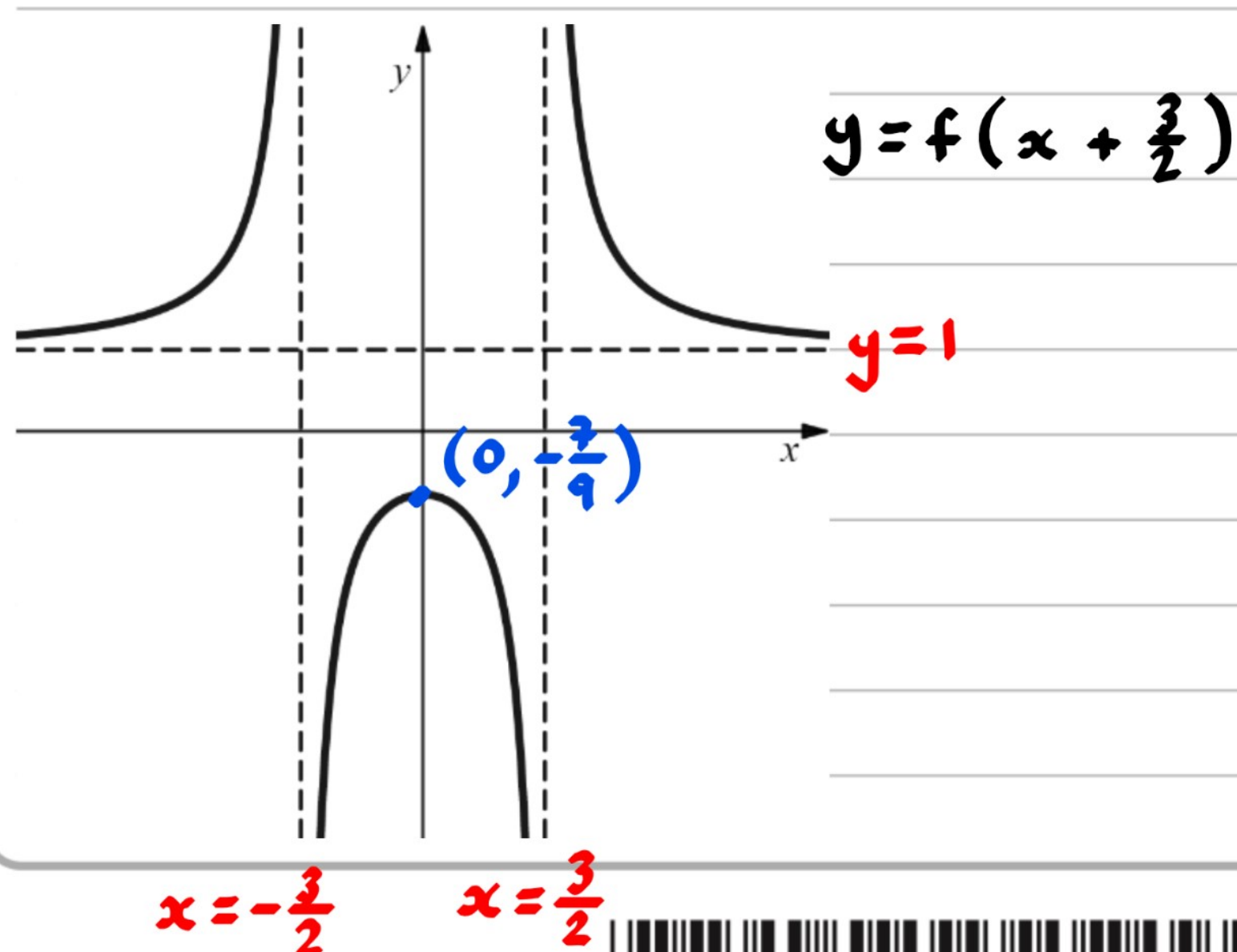
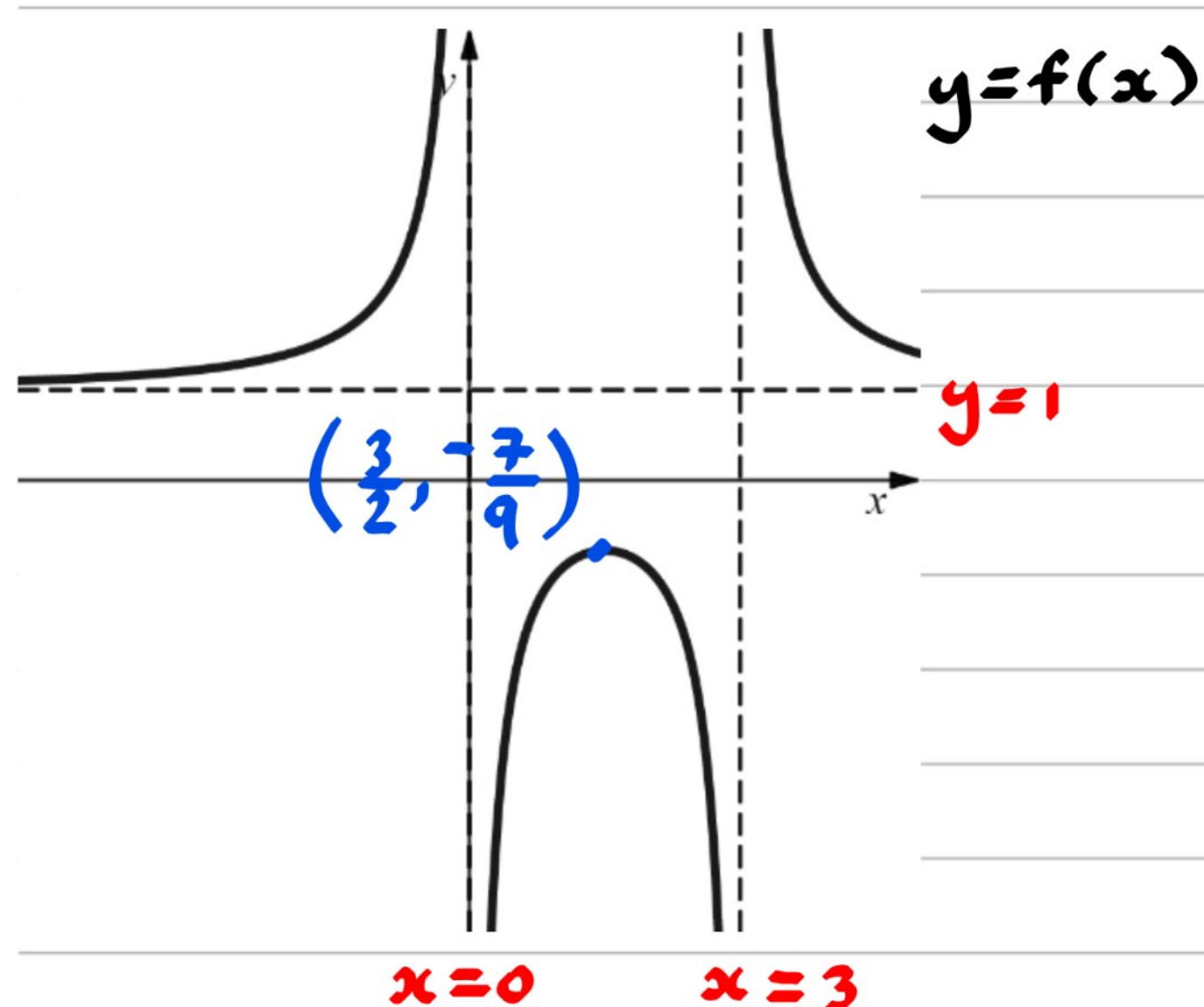
$$f'(x) = 0 : \frac{-4(2x-3)}{(x^2-3x)^2} = 0$$

$$2x - 3 = 0$$

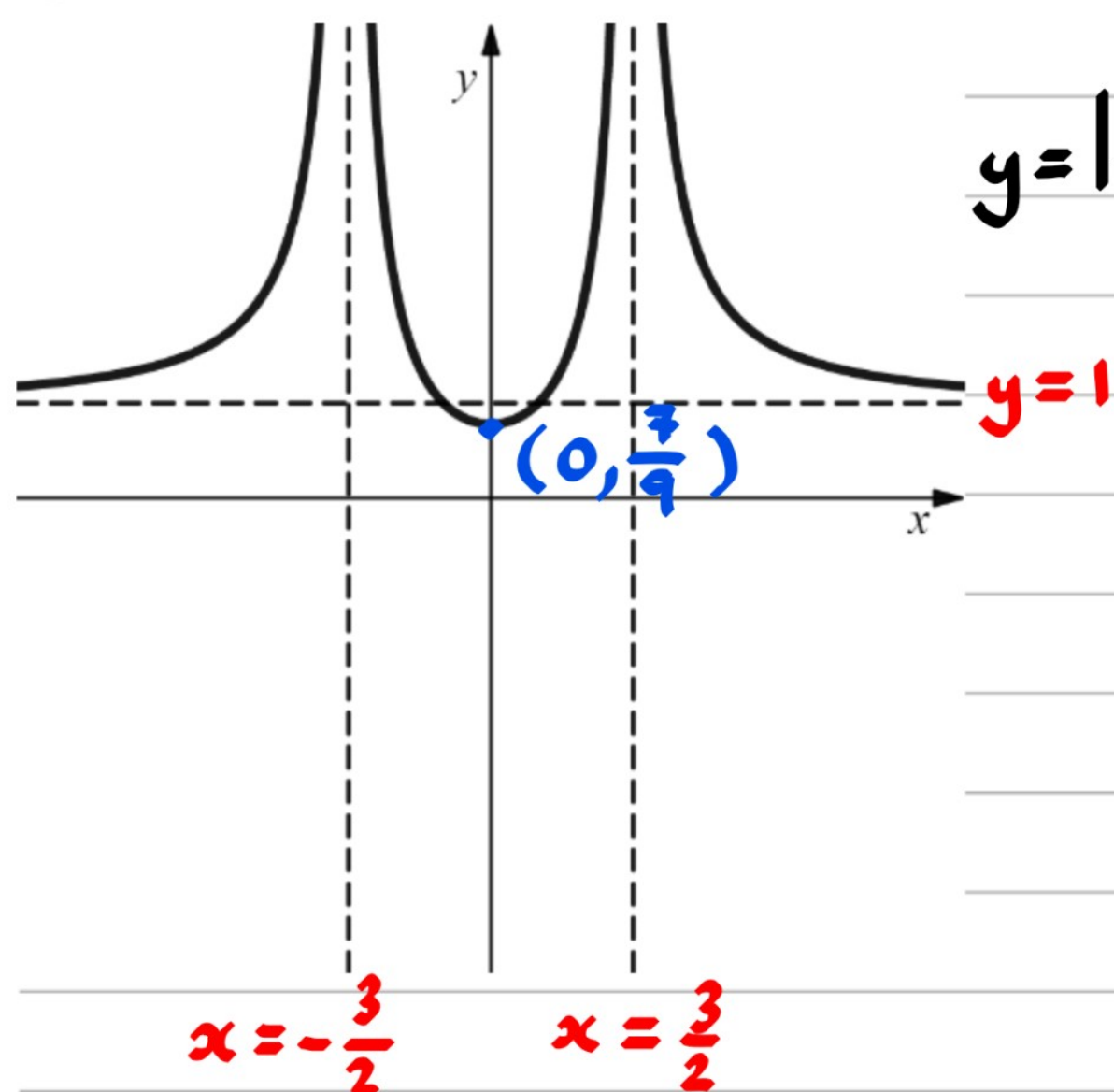
$$\therefore x = \frac{3}{2}$$

$$f\left(\frac{3}{2}\right) = 1 + \frac{4}{\left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right)} = 1 + \frac{4}{\frac{9}{4} - \frac{18}{4}} = 1 - \frac{16}{9} = -\frac{7}{9}$$

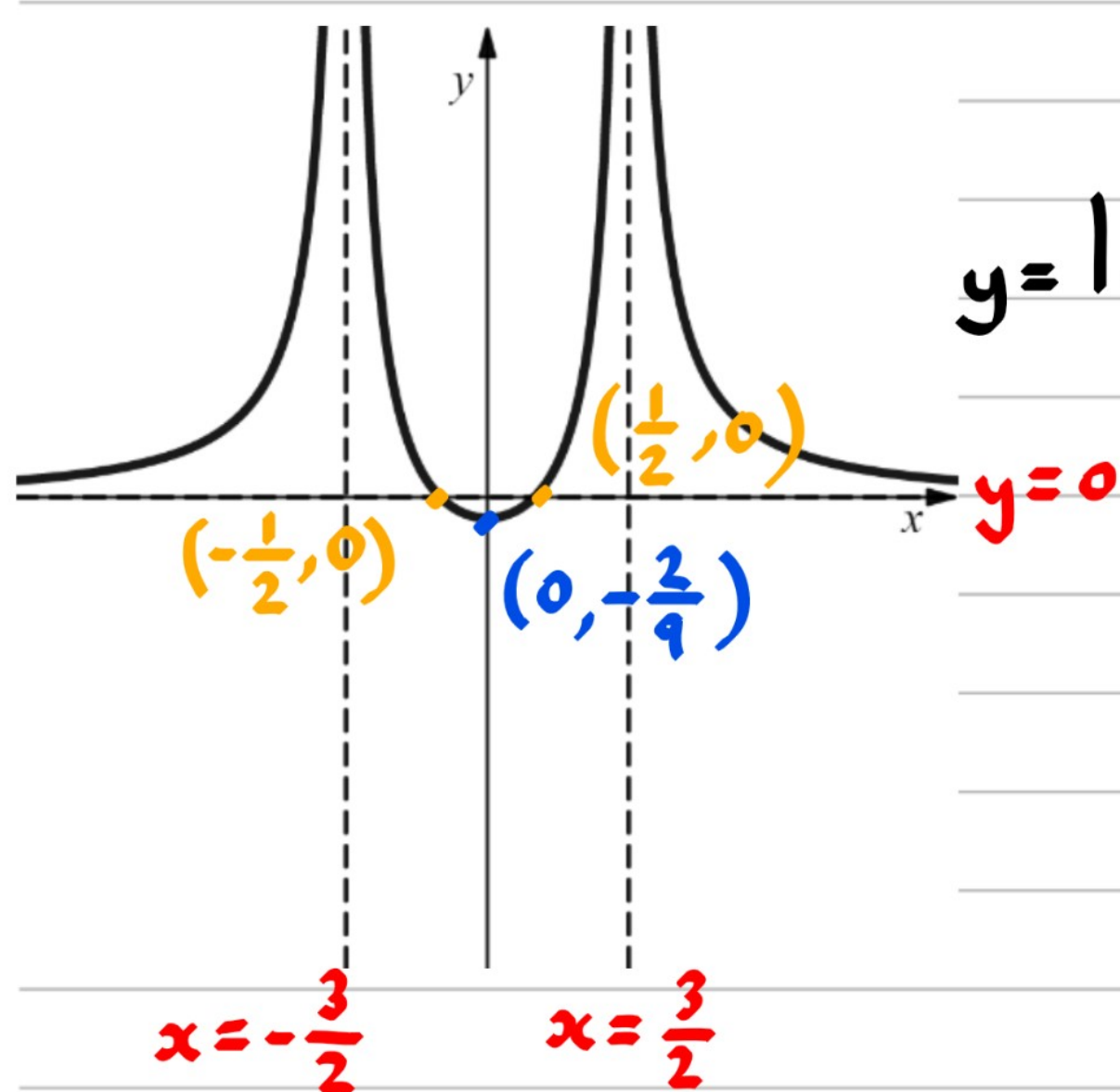
$$\therefore P\left(\frac{3}{2}, -\frac{7}{9}\right)$$



Question 1 continued



$$y = |f(x + \frac{3}{2})|$$



$$y = |f(x + \frac{3}{2})| - 1$$

NOTE: $f(x + \frac{3}{2}) = 1 + \frac{4}{(x + \frac{3}{2})(x + \frac{3}{2} - 3)} = 1 + \frac{4}{(x + \frac{3}{2})(x - \frac{3}{2})}$

i.e. $\left| 1 + \frac{4}{(x + \frac{3}{2})(x - \frac{3}{2})} \right| - 1 = 0$

$$1 + \frac{4}{(x + \frac{3}{2})(x - \frac{3}{2})} = \pm 1$$

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Question 1 continued

$$\oplus \quad 1 + \frac{4}{\left(x + \frac{3}{2}\right)\left(x - \frac{3}{2}\right)} = 1$$

$$\ominus \quad 1 + \frac{4}{\left(x + \frac{3}{2}\right)\left(x - \frac{3}{2}\right)} = -1$$

$$\frac{4}{\left(x + \frac{3}{2}\right)\left(x - \frac{3}{2}\right)} = 0$$

NO SOLUTIONS

$$\left(x + \frac{3}{2}\right)\left(x - \frac{3}{2}\right) = -2$$

$$x^2 - \frac{9}{4} = -2$$

$$4x^2 - 9 = -8$$

$$4x^2 = 1$$

$$x^2 = \frac{1}{4}$$

$$x = \pm \frac{1}{2}$$

$\therefore y = \left|f\left(x + \frac{3}{2}\right)\right| + 1$ INTERSECTS THE x -AXIS AT $\left(\pm \frac{1}{2}, 0\right)$

(Total for Question 1 is 12 marks)



2. The functions f and g are defined by

$$\begin{aligned} f(x) &= 2\sqrt{1 - e^{-x}} & x \in \mathbb{R}, x \geq 0 \\ g(x) &= \ln(4 - x^2) & x \in \mathbb{R}, -2 < x < 2 \end{aligned}$$

(a) (i) Explain why fg cannot be formed as a composite function.

(ii) Explain why gf can be formed as a composite function.

(2)

(b) (i) Find $gf(x)$, giving the answer in the form $gf(x) = a + bx$, where a and b are constants.

(ii) State the domain and range of gf .

(5)

(c) Sketch the graph of the function gf .

On your sketch, you should show the coordinates of any points where the graph meets or crosses the coordinate axes.

(2)

The circle C with centre $(0, -\ln 9)$ touches the line with equation $y = gf(x)$ at precisely one point.

(d) Find an equation of the circle C .

(3)

(+S1)

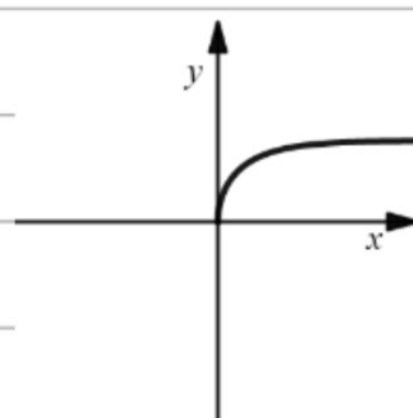
(a) $f(x) = 2\sqrt{1 - e^{-x}}$ DOMAIN: $x \in \mathbb{R}, x \geq 0$

RANGE: $f(0) = 2 \times 0 = 0$

$x \rightarrow \infty, e^{-x} \rightarrow 0 \therefore f(x) \rightarrow 2$

$0 \leq f(x) < 2$

SKETCH:



$g(x) = \ln(4 - x^2)$ DOMAIN: $x \in \mathbb{R}, -2 < x < 2$

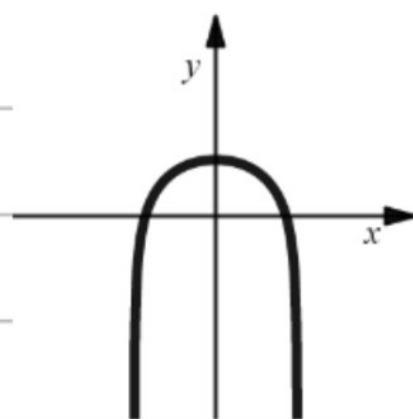
RANGE: $g(0) = \ln 4$

$-\infty < g(x) \leq \ln 4$



Question 2 continued

SKETCH :



(i) RANGE OF g DOES NOT LIE IN DOMAIN OF f

$\therefore fg$ CANNOT BE FORMED

(ii) RANGE OF f LIES IN DOMAIN OF g

$\therefore gf$ CAN BE FORMED

(b) $g(f(x)) = \ln[4 - (2\sqrt{1 - e^{-x}})^2]$

$$= \ln[4 - 4(1 - e^{-x})]$$

$$= \ln(4 - 4 + 4e^{-x})$$

$$= \ln(4e^{-x})$$

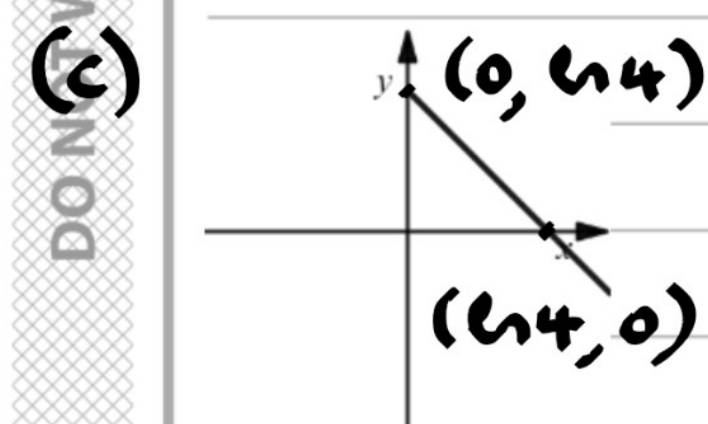
$$= \ln 4 + \ln e^{-x}$$

$$= \ln 4 - x \ln e$$

$$= \ln 4 - x$$

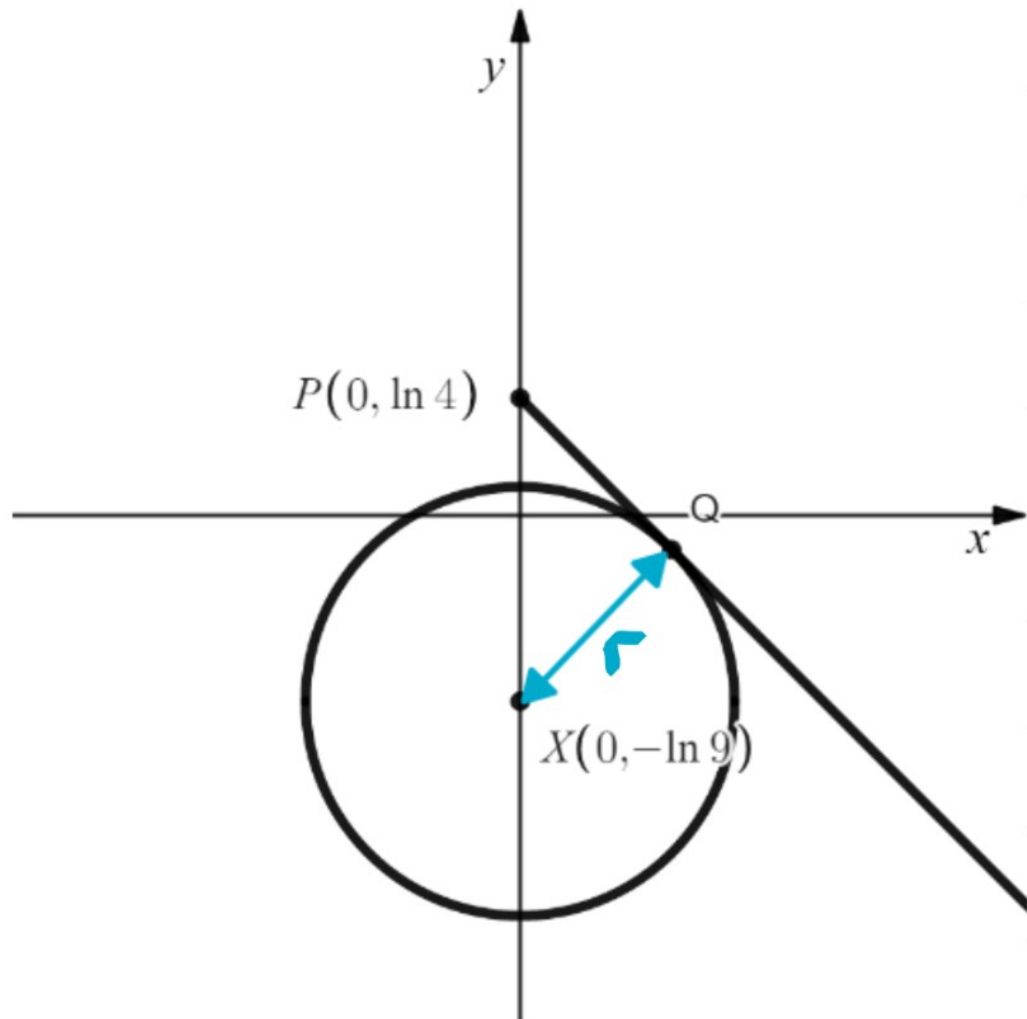
DOMAIN: $x \in \mathbb{R}, x \geq 0$

RANGE: $-\infty < g(f(x)) \leq \ln 4$



Question 2 continued

(a)



$$y = \ln 4 - x \quad \textcircled{1}$$

$$x^2 + (y + \ln 9)^2 = r^2 \quad \textcircled{2}$$

$$\textcircled{1} \rightarrow \textcircled{2}: x^2 + (\ln 4 - x + \ln 9)^2 = r^2$$

$$x^2 + (\ln 36 - x)^2 = r^2$$

$$x^2 + (\ln 36)^2 - 2 \ln 36 x + x^2 = r^2$$

$$2x^2 - 2 \ln 36 x + (\ln 36)^2 - r^2 = 0$$

$$b^2 - 4ac = 0: (-2 \ln 36)^2 - 4(2)\{(\ln 36)^2 - r^2\} = 0$$

$$4(\ln 36)^2 - 8(\ln 36)^2 + 8r^2 = 0$$

$$8r^2 = 4(\ln 36)^2$$

$$r^2 = \frac{1}{2}(\ln 36)^2$$

$$\therefore C: x^2 + (y + \ln 9)^2 = \frac{1}{2}(\ln 36)^2$$



3. (a) (i) Write down the binomial series expansion of

$$\left(1 + \frac{2}{n}\right)^n \quad n \in \mathbb{N}, n > 2$$

in powers of $\left(\frac{2}{n}\right)$ up to and including the term in $\left(\frac{2}{n}\right)^3$

(ii) Hence prove that, for $n \in \mathbb{N}, n \geq 3$

$$\left(1 + \frac{2}{n}\right)^n \geq \frac{19}{3} - \frac{6}{n} \quad (3)$$

(b) Use the binomial series expansion of $\left(1 - \frac{x}{4}\right)^{\frac{1}{2}}$ to show that $\sqrt{3} < \frac{7}{4}$ (4)

$$f(x) = \left(1 + \frac{2}{x}\right)^x - 3^{\frac{x}{6}} \quad x \in \mathbb{R}, x > 0$$

Given that the function $f(x)$ is continuous and that $\sqrt[6]{3} > \frac{6}{5}$

(c) prove that $f(x) = 0$ has a root in the interval $[9, 10]$ (5)

$$\begin{aligned}
 \text{(a)} \quad \left(1 + \frac{2}{n}\right)^n &= 1 + \frac{n}{1!} \left(\frac{2}{n}\right) + \frac{n(n-1)}{2!} \left(\frac{2}{n}\right)^2 + \frac{n(n-1)(n-2)}{3!} \left(\frac{2}{n}\right)^3 + \dots \quad (+S1) \\
 &= 1 + n \left(\frac{2}{n}\right) + \frac{n(n-1)}{2} \left(\frac{2}{n}\right)^2 + \frac{n(n-1)(n-2)}{6} \left(\frac{2}{n}\right)^3 + \dots \\
 &= 1 + 2 + \frac{n(n-1)}{2} \frac{4}{n^2} + \frac{n(n^2-3n+2)}{6} \frac{8}{n^3} + \dots \\
 &= 3 + \frac{2(n-1)}{n} + \frac{4(n^2-3n+2)}{3n^2} + \dots \\
 &= 3 + \frac{2n}{n} - \frac{2}{n} + \frac{4n^2}{3n^2} - \frac{12n}{3n^2} + \frac{8}{3n^2} + \dots \\
 &= 3 + 2 - \frac{2}{n} + \frac{4}{3} - \frac{4}{n} + \frac{8}{3n^2} + \dots
 \end{aligned}$$



Question 3 continued

$$= \frac{19}{3} - \frac{6}{n} + \frac{8}{3n^2} + \dots$$

NOTE: WHEN $n \geq 3$, $\frac{8}{3n^2}$ BECOMES NON-NEGATIVE AND REALLY

SMALL AND $\left(1 + \frac{2}{n}\right)^n$ INCREASES QUICKLY

$$\therefore \left(1 + \frac{2}{n}\right)^n \geq \frac{19}{3} - \frac{6}{n}$$

(6)
$$\left(1 - \frac{x}{4}\right)^{\frac{1}{2}} = 1 + \frac{\left(\frac{1}{2}\right)}{1!} \left(-\frac{x}{4}\right) + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)}{2!} \left(-\frac{x}{4}\right)^2 + \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!} \left(-\frac{x}{4}\right)^3 + \dots$$

$$= 1 + \frac{1}{2} \left(-\frac{x}{4}\right) - \frac{1}{8} \left(\frac{x^2}{16}\right) + \frac{1}{16} \left(\frac{-x^3}{64}\right) + \dots$$

$$= 1 - \frac{x}{8} - \frac{x^2}{128} - \frac{x^3}{1024} + \dots$$

VALID FOR $\left|-\frac{x}{4}\right| < 1 \therefore |x| < 4$

i.e. $\left(1 - \frac{x}{4}\right)^{\frac{1}{2}} < 1 - \frac{x}{8}$ REMOVING ALL NEGATIVE TERMS FROM THE RHS MAKES THIS SIDE LARGE

$$x=1: \left(1 - \frac{1}{4}\right)^{\frac{1}{2}} < 1 - \frac{1}{8}$$

$$\left(\frac{3}{4}\right)^{\frac{1}{2}} < \frac{7}{8}$$

$$\frac{\sqrt{3}}{2} < \frac{7}{8}$$

$$\therefore \sqrt{3} < \frac{7}{4}$$



Question 3 continued

$$(c) f(x) = \left(1 + \frac{2}{x}\right)^x - 3^{\frac{x}{6}}$$

$$f(9) = \left(1 + \frac{2}{9}\right)^9 - 3^{\frac{9}{6}}$$

$$= \left(1 + \frac{2}{9}\right)^9 - \sqrt{3}^3$$

$$= \left(1 + \frac{2}{9}\right)^9 - 3\sqrt{3}$$

$$\text{NOTE : } \left(1 + \frac{2}{9}\right)^9 \geq \frac{19}{3} - \frac{6}{9} \Rightarrow \left(1 + \frac{2}{9}\right)^9 \geq \frac{17}{3}$$

$$\sqrt{3} < \frac{7}{4} \Rightarrow 3\sqrt{3} < \frac{21}{4} \Rightarrow -3\sqrt{3} > -\frac{21}{4}$$

$$\therefore f(9) = \left(1 + \frac{2}{9}\right)^9 - 3\sqrt{3} \geq \frac{17}{3} - \frac{21}{4}$$

$$\left(1 + \frac{2}{9}\right)^9 - 3\sqrt{3} \geq \frac{17 \times 4 - 21 \times 3}{12}$$

$$\left(1 + \frac{2}{9}\right)^9 - 3\sqrt{3} \geq \frac{5}{12}$$

$$\therefore f(9) > 0$$

$$f(10) = \left(1 + \frac{2}{10}\right)^{10} - 3^{\frac{10}{6}}$$

$$= \left(\frac{6}{5}\right)^{10} - (\sqrt[6]{3})^{10}$$



Question 3 continued

NOTE : $\sqrt[6]{3} > \frac{6}{5} \Rightarrow (\sqrt[6]{3})^{10} > \left(\frac{6}{5}\right)^{10} \therefore x^{10}$ IS AN INCREASING FUNCTION

$$\therefore f(10) < 0$$

AS $f(x)$ CHANGES SIGN ON INTERVAL $[9, 10]$ AND IT IS A CONTINUOUS FUNCTION

$\therefore$ THERE IS A ROOT IN THE INTERVAL $[9, 10]$



4.

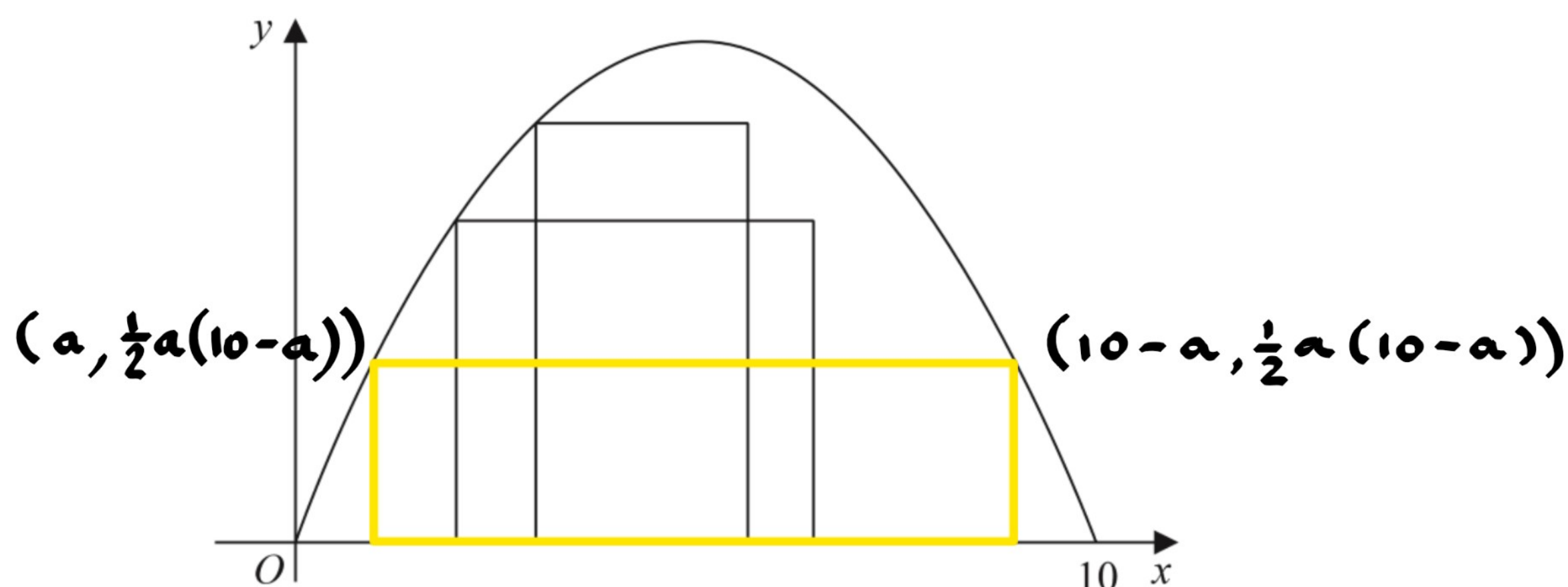


Figure 2

Figure 2 shows a sketch of the parabola with equation $y = \frac{1}{2}x(10 - x)$, $0 \leq x \leq 10$

This question concerns rectangles that lie under the parabola in the first quadrant. The bottom edge of each rectangle lies along the x -axis and the top left vertex lies on the parabola. Some examples are shown in Figure 2.

Let the x coordinate of the top left vertex be a .

(a) Explain why the width, w , of such a rectangle must satisfy $w \leq 10 - 2a$ (2)

(b) Find the value of a that gives the maximum area for such a rectangle. (5)

Given that the rectangle must be a square,

(c) find the value of a that gives the maximum area for such a square. (3)

Given that the area of the rectangles is fixed as 36

(d) find the range of possible values for a . (6)

(+S1)

(a) THE PARABOLA IS SYMMETRICAL ABOUT THE LINE $x = 5$

THE RECTANGLE MUST LIE UNDER THE PARABOLA, SO MAXIMUM WIDTH WILL OCCUR WHEN THE TOP RIGHT VERTEX ALSO LIES ON PARABOLA

THE x -COORDINATE OF THE TOP RIGHT VERTEX IS $10 - a$

THE WIDTH SATISFIES $w \leq 10 - a - a \therefore w \leq 10 - 2a$



Question 4 continued

(b) MAXIMUM AREA MUST OCCUR FOR A FULL WIDTH OF A RECTANGLE

$$\text{i.e. } w = 10 - 2a$$

$$\begin{aligned}\text{MAXIMUM AREA, } A &= \frac{1}{2} a (10 - a)(10 - 2a) \\ &= \frac{1}{2} a (100 - 30a + 2a^2) \\ &= a^3 - 15a^2 + 50a\end{aligned}$$

$$\frac{dA}{da} = 3a^2 - 30a + 50$$

$$\frac{dA}{da} = 0 : 3a^2 - 30a + 50 = 0$$

$$3[a^2 - 10a] + 50 = 0$$

$$3[(a - 5)^2 - 25] = 0$$

$$(a - 5)^2 = \frac{25}{3}$$

$$a = 5 \pm \frac{5}{\sqrt{3}}$$

AS $0 < a < 5$ $\therefore$ NEED VALID RECTANGLE AND AREA IS ZERO AT EITHER END OF THIS INTERVAL

$$\therefore \text{MAXIMUM AREA OCCURS WHEN } a = 5 - \frac{5}{\sqrt{3}}$$

(c) FOR SQUARE: $10 - 2a = \frac{1}{2} a (10 - 2a)$

$$20 - 4a = 10a - a^2$$



Question 4 continued

$$a^2 - 14a + 20 = 0$$

$$(a - 7)^2 - 49 + 20 = 0$$

$$(a - 7)^2 = 29$$

$$a = 7 \pm \sqrt{29}$$

$$\text{AS } 0 < a < 5 \text{ AND } \sqrt{29} < 7$$

$$\therefore a = 7 - \sqrt{29}$$

$$(d) \text{ IF AREA IS } 36 : w = \frac{36}{\frac{1}{2}a(10-a)} = \frac{72}{a(10-a)}$$

$$\text{AS } w \leq 10 - 2a : \frac{72}{a(10-a)} \leq 10 - 2a$$

$$72 \leq a(10-a)(10-2a)$$

$$72 \leq a(100 - 30a + 2a^2)$$

$$72 \leq 100a - 30a^2 + 2a^3$$

$$2a^3 - 30a^2 + 100a - 72 \geq 0$$

$$\therefore a^3 - 15a^2 + 50a - 36 \geq 0$$

$$\text{LET } f(a) = a^3 - 15a^2 + 50a - 36$$

$$\text{FACTORS OF } 36 : \pm \{1, 36, 2, 18, 3, 12, 4, 9, 6\}$$

$$\text{TRY: } f(1) = (1)^3 - 15(1)^2 + 50(1) - 36 = 1 - 15 + 50 - 36 = 0$$

$$\therefore (a - 1) \text{ IS A FACTOR OF } f(a)$$

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Question 4 continued

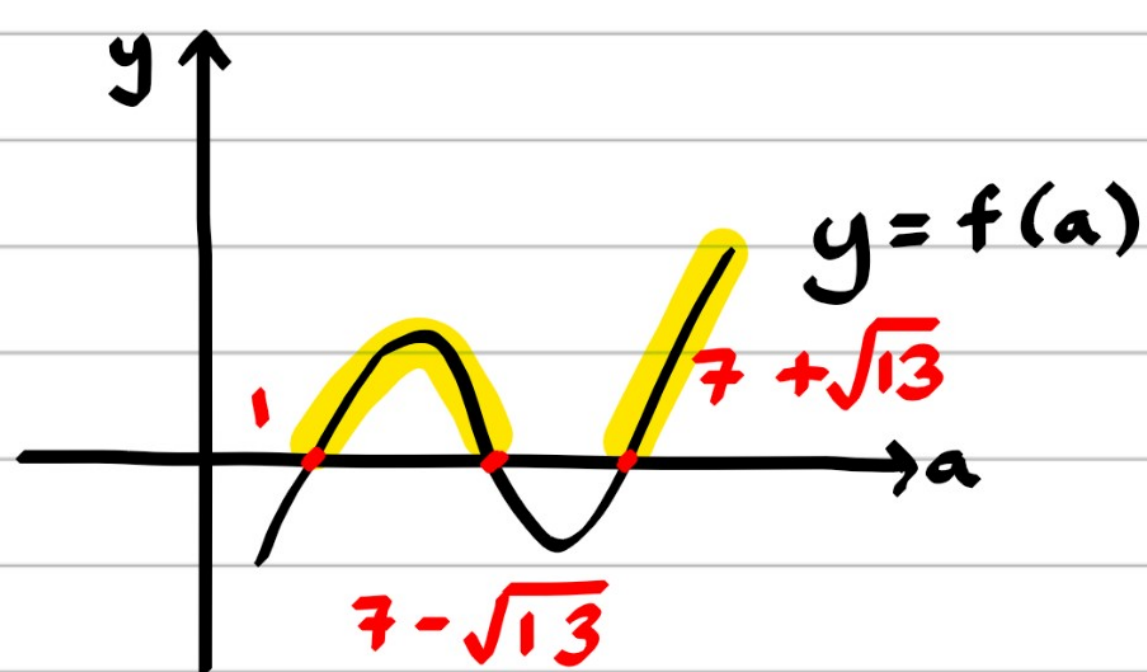
$$\therefore (a-1)(a^2-14a+36) \equiv a^3-15a^2+50a-36$$

$$\therefore f(a)=0: (a-1)(a^2-14a+36)=0$$

$$(a-1)[(a-7)^2-49+36]=0$$

$$(a-1)[(a-7)^2-13]=0$$

$$\therefore \text{cv: } a=1, 7 \pm \sqrt{13}$$



$$\therefore 1 \leq a \leq 7 - \sqrt{13}, a \geq 7 + \sqrt{13}$$

$$\text{As } 0 < a < 5$$

$$\therefore 1 \leq a \leq 7 - \sqrt{13}$$



5. (a) The box below shows a student's attempt to prove the following identity for $a > b > 0$

$$\arctan a - \arctan b \equiv \arctan \frac{a-b}{1+ab}$$

Let $x = \arctan a$ and $y = \arctan b$, so that $a = \tan x$ and $b = \tan y$

$$\text{So } \tan(\arctan a - \arctan b) \equiv \tan(x - y)$$

$$\begin{aligned} &\equiv \frac{\tan x - \tan y}{1 - \tan^2(xy)} \\ &\equiv \frac{a - b}{1 - (ab)^2} \\ &\equiv \frac{a - ab + ab - b}{(1 - ab)(1 + ab)} \\ &\equiv \frac{a(1 - ab) - b(1 - ab)}{(1 - ab)(1 + ab)} \\ &\equiv \frac{a - b}{1 + ab} \end{aligned}$$

Taking arctan of both sides gives $\arctan a - \arctan b \equiv \arctan \frac{a-b}{1+ab}$ as required.

There are three errors in the proof where the working does not follow from the previous line.

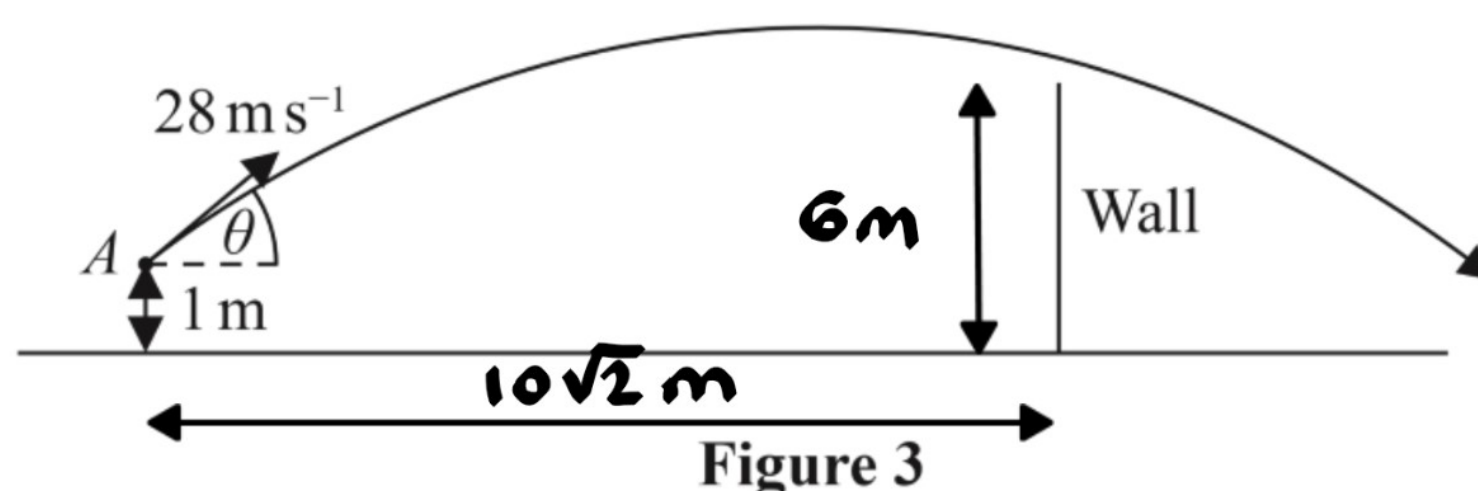
- (i) Describe these three errors.

(3)

- (ii) Write out a correct proof of the identity.

(2)

- (b) [In this question take g to be 9.8 m s^{-2}]



Balls are projected, one after another, from a point, A , one metre above horizontal ground. Each ball travels in a vertical plane towards a 6 metre high vertical wall of negligible thickness, which is a horizontal distance of $10\sqrt{2}$ metres from A .

The balls are modelled as particles and it is assumed that there is no air resistance.

Each ball is projected with an initial speed of 28 m s^{-1} and at a random angle θ to the horizontal, where $0 < \theta < 90^\circ$



Question 5 continued

Given that a ball will pass over the wall precisely when $\alpha \leq \theta \leq \beta$

- (i) find, in degrees, the angle $\beta - \alpha$

(10)

- (ii) Deduce that the probability that a particular ball will pass over the wall is $\frac{2}{3}$

(1)

- (iii) Hence find the probability that exactly 2 of the first 10 balls projected pass over the wall.

You should give your answer in the form $\frac{P}{Q^k}$ where P , Q and k are integers and P is not a multiple of Q .

(3)

- (iv) Explain whether taking air resistance into account would increase or decrease the probability in (b)(iii).

(1)

(+S2)

(a)(i) THE EXPANSION OF $\tan(x-y)$ IS INCORRECT BETWEEN LINES 2 AND 3, THE DENOMINATOR SHOULD BE $1 + \tan x \tan y$

BETWEEN LINES 3 AND 4, WHEN REPLACING TANGENTS, $\tan^2 xy$ IS NOT $(ab)^2$; THE STUDENT HAS INCORRECTLY ASSUMED $(\tan xy)^2 = (\tan x \tan y)^2$

THE FACTORISATION BETWEEN LINES 5 AND 6 IS INCORRECT:
 $a - ab \equiv a(1-b)$, NOT $a(1-ab)$

(ii) $\tan^{-1} a - \tan^{-1} b \equiv \tan^{-1} \left(\frac{a-b}{1+ab} \right)$

LET $x = \tan^{-1} a$ AND $y = \tan^{-1} b$ SO $\tan x = a$ AND $\tan y = b$

SO: $\tan(\tan^{-1} a - \tan^{-1} b) = \tan(x-y)$

$$= \frac{\tan x - \tan y}{1 + \tan x \tan y}$$



$$= \frac{a-b}{1+ab}$$

$$\therefore \tan^{-1} a - \tan^{-1} b = \tan^{-1} \left(\frac{a-b}{1+ab} \right)$$

(b)(i) HORIZONTAL: $S_x = 28 \cos \theta \times t$

$$10\sqrt{2} = 28 \cos \theta \times t$$

$$\therefore t = \frac{10\sqrt{2}}{28 \cos \theta}$$

VERTICAL: $S_y = 28 \sin \theta t - \frac{1}{2} g t^2$ (RELATIVE TO A)

$$S_y > 5 : 28 \sin \theta t - \frac{1}{2} g t^2 > 5$$

$$28 \sin \theta \left(\frac{10\sqrt{2}}{28 \cos \theta} \right) - \frac{1}{2} g \left(\frac{10\sqrt{2}}{28 \cos \theta} \right)^2 > 5$$

$$\frac{\sin \theta}{\cos \theta} \cdot 10\sqrt{2} - \frac{9.8}{2} \cdot \frac{200}{28 \times 28} \cdot \frac{1}{\cos^2 \theta} > 5$$

$$10\sqrt{2} \tan \theta - \frac{49}{10} \cdot \frac{25}{98} \cdot \sec^2 \theta > 5$$

$$10\sqrt{2} \tan \theta - \frac{5}{4} \sec^2 \theta > 5$$

$$2\sqrt{2} \tan \theta - \frac{1}{4} \sec^2 \theta > 1$$

$$8\sqrt{2} \tan \theta - \sec^2 \theta > 4$$

$$8\sqrt{2} \tan \theta - (1 + \tan^2 \theta) > 4$$

$$\tan^2 \theta - 8\sqrt{2} \tan \theta + 5 > 0$$



Question 5 continued

$$(\tan \theta - 4\sqrt{2})^2 - 32 + 5 > 0$$

$$(\tan \theta - 4\sqrt{2})^2 > 27$$

$$\text{C.V. : } \tan \theta = 4\sqrt{2} \pm 3\sqrt{3}$$

$$\text{i.e. } \beta - \alpha = \tan^{-1} \left(\frac{(4\sqrt{2} + 3\sqrt{3}) - (4\sqrt{2} - 3\sqrt{3})}{1 + (4\sqrt{2} + 3\sqrt{3})(4\sqrt{2} - 3\sqrt{3})} \right)$$

$$= \tan^{-1} \left(\frac{6\sqrt{3}}{1 + (32 - 27)} \right)$$

$$= \tan^{-1} \left(\frac{6\sqrt{3}}{6} \right)$$

$$= \tan^{-1}(\sqrt{3})$$

$$= 60^\circ$$

(ii) PROBABILITY THAT ANY GIVEN BALL SUCCESSFULLY CLEARS THE WALL IS GIVEN BY

$$p = \frac{60^\circ}{90^\circ} = \frac{2}{3}$$

(iii) LET R.V. $X = \text{NO}^\circ \text{ OF BALLS PASSING OVER WALL OUT OF 10 ATTEMPTS}$

$$\text{THEN } X \sim B \left(10, \frac{2}{3} \right)$$

$$\therefore P(X = 2) = {}^{10}C_2 \left(\frac{2}{3} \right)^2 \left(\frac{1}{3} \right)^8$$

$$= \frac{10!}{2!(10-2)!} \left(\frac{2}{3} \right)^2 \left(\frac{1}{3} \right)^8$$



Question 5 continued

$$= \frac{10 \times 9 \times 8!}{2! 8!} \cdot \frac{4}{9} \cdot \frac{1}{3^8}$$

$$= \frac{10 \times 9 \times 4}{2 \times 9 \times 3^8}$$

$$= \frac{20}{3^8}$$

(iv) TAKING AIR RESISTANCE INTO ACCOUNT WOULD DECREASE RANGE, SO THE PROBABILITY OF SUCCESS WOULD BE REDUCED $\therefore$ PROBABILITY OF TWO SUCCESSES WOULD INCREASE

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6. (a) Given that f is a function such that the integrals exist,

(i) use the substitution $u = a - x$ to show that

$$\int_0^a f(x) dx = \int_0^a f(a - x) dx \quad (2)$$

(ii) Hence use symmetry of $f(\sin x)$ on the interval $[0, \pi]$ to show that

$$\int_0^\pi x f(\sin x) dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx \quad (4)$$

(b) Use the result of (a)(i) to show that

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx$$

is independent of n , and find the value of this integral.

(4)

(c) (i) Prove that

$$\frac{\cos x}{1 + \cos x} \equiv 1 - \frac{1}{2} \sec^2 \left(\frac{x}{2} \right)$$

(ii) Hence use the results from (a) to find

$$\int_0^\pi \frac{x \sin x}{1 + \sin x} dx \quad (7)$$

(d) Find

$$\int_0^\pi \frac{x \sin^4 x}{\sin^4 x + \cos^4 x} dx \quad (4)$$

(+S2)

(a)(i) $u = a - x \quad \therefore x = a - u$

$\frac{dx}{du} = -1 \quad \therefore dx = -du$

$x = 0 : u = a$



Question 6 continued

$$x = a : u = 0$$

$$\therefore \int_0^a f(x) dx \text{ TRANSFORMS TO } \int_a^0 f(a-u) \cdot -du \equiv \int_0^a f(a-u) du$$

NOTE: AS IT IS A DEFINITE INTEGRAL YOU CAN USE ANY VARIABLE YOU LIKE; IT'S GOING TO GIVE YOU AN ANSWER (IN TERMS OF a), AND THAT NUMBER IS THE SAME FOR WHATEVER YOU CALLED THE VARIABLE

SO YOU CAN REPLACE u WITH x

$$\therefore \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$(ii) \int_0^\pi x f(\sin x) dx = \int_0^\pi (\pi - x) f(\sin(\pi - x)) dx$$

$$\int_0^\pi x f(\sin x) dx = \int_0^\pi \pi f(\sin(\pi - x)) dx - \int_0^\pi x f(\sin(\pi - x)) dx$$

NOTE: $\sin x \equiv \sin(\pi - x)$ FOR $x \in \mathbb{R}$

$$\int_0^\pi x f(\sin x) dx = \int_0^\pi \pi f(\sin x) dx - \int_0^\pi x f(\sin(\pi - x)) dx$$

$$2 \int_0^\pi x f(\sin x) dx = \int_0^\pi \pi f(\sin x) dx$$

NOTE: $f(\sin x)$ IS SYMMETRIC ABOUT THE LINE WITH EQUATION $x = \frac{\pi}{2}$ ON THE INTERVAL $[0, \pi]$

$$2 \int_0^\pi x f(\sin x) dx = 2 \int_0^{\frac{\pi}{2}} \pi f(\sin x) dx$$

$$\therefore \int_0^\pi x f(\sin x) dx = \int_0^{\frac{\pi}{2}} \pi f(\sin x) dx$$

(b) WANT TO SHOW $\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = A$, FOR SOME CONSTANT A

AND THIS IS INDEPENDENT OF n



Question 6 continued

$$\text{using (a)(i)} : \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\sin^n(\frac{\pi}{2} - x)}{\sin^n(\frac{\pi}{2} - x) + \cos^n(\frac{\pi}{2} - x)} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\cos^n x + \sin^n x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^n x}{\sin^n x + \cos^n x} dx + \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx$$

$$2 \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^n x + \sin^n x}{\sin^n x + \cos^n x} dx$$

$$2 \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\frac{\pi}{2}} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{1}{2} [x]_0^{\frac{\pi}{2}}$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{1}{2} \left\{ \frac{\pi}{2} \right\} - \frac{1}{2} \{0\}$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \frac{\pi}{4} \text{ FOR ANY } n$$

$\therefore$ INTEGRAL IS INDEPENDENT OF n AND ITS VALUE IS $\frac{\pi}{4}$

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Question 6 continued

$$\begin{aligned}
 (c)(i) \quad \frac{\cos x}{1 + \cos x} &\equiv \frac{1 + \cos x - 1}{1 + \cos x} \\
 &\equiv \frac{1 + \cos x}{1 + \cos x} - \frac{1}{1 + \cos x} \\
 &\equiv 1 - \frac{1}{1 + \cos x} \\
 &\equiv 1 - \frac{1}{1 + \{2\cos^2(\frac{x}{2}) - 1\}} \\
 &\equiv 1 - \frac{1}{2\cos^2(\frac{x}{2})} \\
 &\equiv 1 - \frac{1}{2} \sec^2\left(\frac{x}{2}\right) \\
 &\equiv \text{RHS, HENCE SHOWN}
 \end{aligned}$$

$$(ii) \text{ USING (a)(ii): } \int_0^\pi \frac{x \sin x}{1 + \sin x} dx = \pi \int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x} dx$$

$$\text{USING (a)(i): } \pi \int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x} dx = \pi \int_0^{\frac{\pi}{2}} \frac{\sin(\frac{\pi}{2} - x)}{1 + \sin(\frac{\pi}{2} - x)} dx$$

$$\therefore \int_0^\pi \frac{x \sin x}{1 + \sin x} dx = \pi \int_0^{\frac{\pi}{2}} \frac{\sin(\frac{\pi}{2} - x)}{1 + \sin(\frac{\pi}{2} - x)} dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \cos x} dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \left\{ 1 - \frac{1}{2} \sec^2\left(\frac{x}{2}\right) \right\} dx \quad (\text{using (c)(i)})$$

$$= \pi \left[x - \tan\left(\frac{x}{2}\right) \right]_0^{\frac{\pi}{2}}$$



Question 6 continued

$$= \pi \left\{ \frac{\pi}{2} - \tan\left(\frac{\pi}{4}\right) \right\}$$

$$= \pi \left\{ \frac{\pi}{2} - 1 \right\}$$

$$= \frac{\pi^2}{2} - \pi$$

(d) NOTE: WANT TO GET INTEGRAL IN TERMS OF $\sin x$ ONLY TO APPLY PREVIOUS KNOWN RESULTS

$$\int_0^{\pi} \frac{x \sin^4 x}{\sin^4 x + \cos^4 x} dx \equiv \int_0^{\pi} \frac{x \sin^4 x}{\sin^4 x + (\cos^2 x)^2} dx$$

$$\equiv \int_0^{\pi} \frac{x \sin^4 x}{\sin^4 x + (1 - \sin^2 x)^2} dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{\sin^4 x}{\sin^4 x + (1 - \sin^2 x)^2} dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{\sin^4 x}{\sin^4 x + (\cos^2 x)^2} dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{\sin^4 x}{\sin^4 x + \cos^4 x} dx$$

$$= \pi \left(\frac{\pi}{4} \right)$$

$$= \frac{\pi^2}{4}$$

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